Richness-Aware Modality-Role Assignment for Multimodal Emotional Dialogue Generation

Mingting Yu[0009-0007-4247-0747], Changhong Jiang[0000-0002-2647-3987]*, Zhongtian Hu[0000-0001-5349-4751], Wei Zhang[0000-0003-2951-4833], and Lulu Wang[0009-0001-8052-5987]

Northwestern Polytechnical University, Xi'an 710129, China
mingtingyu@mail.nwpu.edu.cn, chjiang@nwpu.edu.cn, ahxchzt@mail.nwpu.edu.cn, zhang_wei@mail.nwpu.edu.cn, wangllu@mail.nwpu.edu.cn

Abstract. Multimodal emotional dialogue generation (EDG) has benefited from increasingly powerful fusion architectures, yet many systems still adopt a text-centric and statically fused multimodal pipeline and fail to fully exploit audio–visual affective cues whose salience varies across dialogue contexts. We present **RaMRA**, a *richness-aware modality-role assignment* framework that estimates the semantic–sentiment informativeness of text, audio, and vision and dynamically assigns instance-level dominant/auxiliary roles. The dominant stream serves as an interaction hub for role-conditioned cross-modal attention, while response generation remains anchored in the textual context. Experiments on IEMOCAP and MELD demonstrate that RaMRA consistently improves fluency, semantic alignment, and emotion consistency over strong baselines, underscoring the importance of adaptive multimodal affect perception for building emotionally intelligent dialogue agents.

Keywords: Multimodal dialogue generation · Modality-role assignment · Role-aware fusion · Affective computing · Emotional intelligence

1 Introduction

Dialogue generation has become a core interface for intelligent customer service, virtual assistants, and everyday human–computer interaction. While neural generators increasingly master coherence and contextual relevance, they still tend to under-model affect, resulting in responses that sound fluent yet emotionally off-target [8]. Emotional dialogue generation (EDG) therefore seeks to produce replies that are not only contextually appropriate but also aligned with the speaker's affective state—a capability at the heart of *emotional intelligence* in human–AI interaction and a prerequisite for truly empathetic dialogue agents.

* Corresponding author.

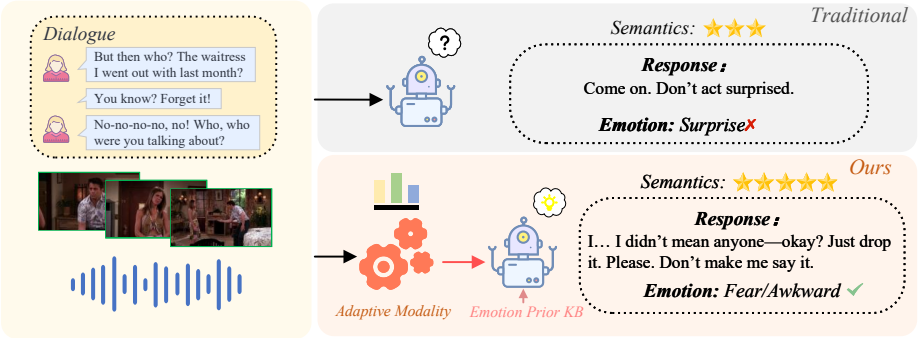


Fig. 1. Motivation and overview of **RaMRA**. Conventional multimodal EDG is often text-centric, which can overlook context-dependent affective cues. The Modality-Role Discriminator and Role-Aware Fusion Module jointly assign primary/auxiliary roles and route fusion through the primary modality, while an *Emotion Prior KB* injects affective priors to improve emotion-aligned generation.

Multimodal EDG is often motivated by a simple observation: the same words can convey different emotions under different prosody or facial display. Recent systems accordingly incorporate audio and vision [3, 6], yet many pipelines remain *text-centric* in a structural sense—text acts as the default backbone, and non-text streams are appended as auxiliary refinements or global conditions. This design implicitly assumes that extra modalities are uniformly helpful, which is rarely true in practice: acoustic cues may dominate when wording is neutral but tone is tense, while facial expressions can be decisive when speech is short or ambiguous. Even systems built around a *fixed* non-text primary modality inherit the same limitation—the weighting prior is predetermined rather than inferred per instance, making them brittle under modality-specific noise, missing cues, or cross-modal incongruence [22, 12, 6]. This motivates a key question: "can multimodal EDG explicitly decide which modality should lead the interaction for each dialogue context, while keeping generation anchored to textual discourse?"

As illustrated in Fig. 1, we argue that multimodal EDG should separate two functional roles: (i) a *grounding channel* that preserves dialogue semantics and controllability (naturally, text), and (ii) an *interaction hub* that adaptively routes cross-modal affective evidence to maximize emotional reliability. To instantiate this principle, we propose **RaMRA**[1], an informativeness-driven modality-role assignment and role-conditioned interaction framework within a decoder-only generation pipeline. For each dialogue context, RaMRA estimates the semantic–sentiment informativeness of text, audio, and vision, assigns a dominant/auxiliary role configuration, and performs cross-modal exchange *asymmetrically*: the dominant modality serves as the sole query source for cross-modal attention, while auxiliary streams contribute evidence only through value vectors modulated by their informativeness weights. This role-based asymmetry dis-

[1] RaMRA stands for **R**ichness-**a**ware **M**odality-**R**ole **A**ssignment.

tinguishes RaMRA from prior dynamic fusion approaches [21, 15, 24, 14], which reweight modality contributions through *scalar* gates yet still maintain symmetric participation of every modality in attention. In RaMRA, high-quality signals are amplified through hub-based routing rather than averaged with noise from less reliable streams. Decoding remains anchored in textual discourse, and an auxiliary emotion prediction objective on decoder hidden states further tightens the coupling between affective understanding and response generation [8].

Our contributions are threefold:

1. **Role-based reformulation.** We formulate multimodal EDG as *instance-level modality-role assignment* and propose a richness-aware criterion that assigns roles based on semantic–sentiment informativeness.
2. **Role-based asymmetric fusion.** We introduce role-conditioned asymmetric interaction that routes cross-modal evidence via gated attention, reducing redundancy and interference from less informative streams.
3. **Framework & validation.** We present RaMRA as a plug-in framework on top of a decoder-only generator with frozen modality encoders, and validate its effectiveness on IEMOCAP and MELD through analyses and ablations.

2 Related Work

Emotional Dialogue Generation. Emotional dialogue generation (EDG) aims to produce responses that are both contextually coherent and emotionally aligned with the speaker's affective state. Text-only approaches have progressively enriched affective grounding through commonsense knowledge infusion [9], emotion-sensitive recognition [13], coarse-to-fine cognition–affection alignment [27], and iterative associative memory [23]. A closely related upstream task, multimodal emotion recognition in conversations (ERC), provides the perceptual foundation that EDG systems rely on; recent ERC advances include attention-based correlation-aware fusion [10], adaptive graph learning [11], and graph-spectral analysis of cross-modal emotion dynamics [1], as surveyed in [16].

Building on these recognition capabilities, multimodal EDG systems have begun to incorporate audio and visual signals alongside text. Representative efforts include the open-source avatar chatbot EmpathyEar [3], the emotion-guided LLM pipeline SpeakFromHeart [6], multi-modality fusion in ERGM [18], LLM-driven empathetic speech generation in EmpathyOmni [12], the text-speech-vision avatar benchmark AvaMERG [25], and personality-aware multimodal empathetic generation [17]. Despite their progress, these systems share a common architectural assumption: either text serves as the fixed backbone with non-text streams appended as auxiliary signals, or a single non-text modality is predesignated as the primary channel. Neither design adapts to the *instance-level* variation in modality salience that naturally arises in emotional conversations.

Dynamic and Non-uniform Multimodal Fusion. Standard multimodal fusion strategies—such as concatenation, element-wise gating, and symmetric cross-attention—treat all modalities as equally informative, which can amplify noise

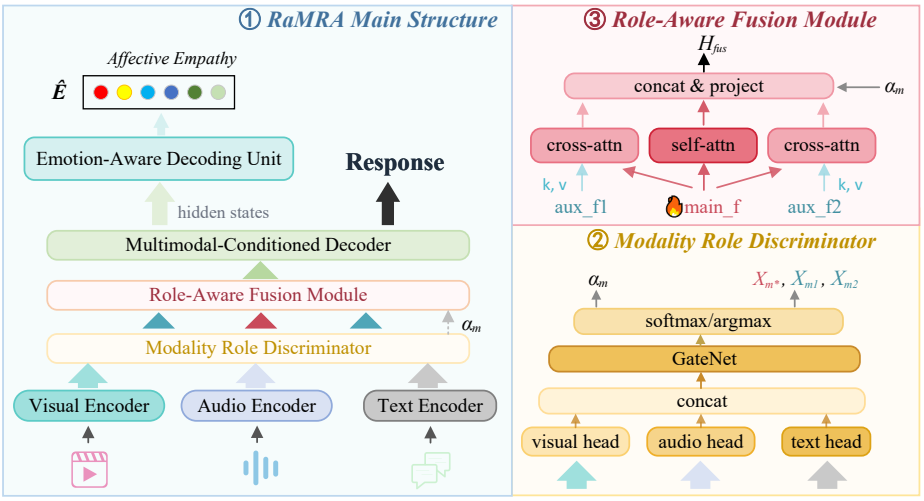


Fig. 2. Architecture of **RaMRA**, organized in three panels read in the order (1)→(2)→(3) to mirror the data flow. **(1) RaMRA Main Structure** (left): the end-to-end pipeline where frozen encoders feed the Modality-Role Discriminator (MRD), whose role weights α_m drive the Role-Aware Fusion Module (RAF), and an Emotion-Aware Decoding Unit supplies auxiliary affective supervision on decoder hidden states. Panels (2) and (3) zoom into the two highlighted submodules of (1): **(2) MRD** (bottom-right) produces α_m via gated tri-tower scoring, and **(3) RAF** (top-right) performs asymmetric routing where the dominant stream serves as the sole query source for cross-attention over auxiliaries, yielding $\mathbf{H}_{\text{fus}}$ for the decoder in (1).

from unreliable streams and dilute discriminative affective cues [4]. To address this, Dynamic Multimodal Fusion [21] estimates per-modality reliability at test time and adjusts contribution weights accordingly; HCMS [15] introduces hierarchical and conditional modality selection for efficient video recognition; and pre-gating with contextual attention gating [24] provides learnable mechanisms to suppress noisy channels before fusion. More recent work explores adaptive modality weighting through multimodal transformers [14] and multi-loss optimization across heterogeneous feature encoders [19], both targeting sentiment analysis. While these methods adaptively reweight modality contributions, they typically apply scalar adjustments that keep every modality symmetrically engaged in fusion, which is precisely the limitation RaMRA addresses through explicit dominant/auxiliary role assignment and asymmetric, role-conditioned routing (as outlined in Sec. 1).

3 Methodology

Task Definition. We study multimodal emotional dialogue generation with aligned text, audio, and vision. Given a dialogue context, we generate a contextually appropriate and emotion-aligned response; emotion labels are used only for training

supervision and are not available at inference. Fig. 2 overviews RaMRA, which assigns modality roles to route cross-modal evidence while keeping decoding grounded in text.

3.1 Modality-Role Discriminator

Let $\mathcal{M} = \{T, A, V\}$ denote *text*, *audio*, and *vision*. Given a dialogue context, modality-specific encoders map raw inputs to modality-level embeddings:

$$\mathbf{x}_m = E_m(\text{input}_m), \quad m \in \mathcal{M}, \tag{1}$$

where E_T, E_A, E_V are fixed feature extractors. RaMRA operates exclusively on $\{\mathbf{x}_m\}$, avoiding direct dependence on raw waveforms or frames.

Tri-Tower Richness Scoring with Gated Calibration. We adopt a tri-tower parameterization to compute modality-specific latent summaries:

$$\mathbf{h}_m = \phi\big(\text{LN}(\mathbf{W}_m^{(h)}\mathbf{x}_m + \mathbf{b}_m^{(h)})\big), \quad m \in \mathcal{M}, \tag{2}$$

where $\phi(\cdot)$ is a nonlinearity and $\text{LN}(\cdot)$ stabilizes representation scales. To suppress unreliable dimensions, we introduce a learnable multiplicative gate:

$$\mathbf{g}_m = \sigma\big(\mathbf{W}_m^{(g)}\mathbf{x}_m + \mathbf{b}_m^{(g)}\big), \tag{3}$$

$$\tilde{\mathbf{h}}_m = \mathbf{g}_m \odot \mathbf{h}_m, \tag{4}$$

with sigmoid $\sigma(\cdot)$ and Hadamard product $\odot$. We then derive a scalar informativeness score via an affine readout:

$$s_m = \mathbf{u}_m^\top \tilde{\mathbf{h}}_m + c_m, \tag{5}$$

and convert scores into a continuous role distribution using a temperature-scaled normalization:

$$\alpha_m = \frac{\exp(s_m/\tau)}{\sum_{m' \in \mathcal{M}} \exp(s_{m'}/\tau)}, \quad m \in \mathcal{M}, \tag{6}$$

where τ controls the sharpness of the role distribution.

We define *informativeness* (or *richness*) operationally as the expected contribution of a modality to the joint objectives of semantic coherence and sentiment prediction in the current dialogue context. Concretely, for each modality m, the gated latent summary $\tilde{\mathbf{h}}_m$ (Eq. 4) isolates dimensions that carry discriminative affective and semantic value by multiplicatively suppressing uninformative channels; the scalar score s_m (Eq. 5) then aggregates this filtered signal into a single informativeness estimate. The entire pathway—from modality encoding through gating to scoring—is learned end-to-end under the combined language modeling and emotion classification losses (Eq. 15), so that a high s_m reflects evidence that is simultaneously useful for generating fluent, coherent text and for predicting the speaker's emotion. This joint optimization ensures that informativeness is

not a hand-crafted heuristic but an empirically grounded quantity aligned with the downstream task.

We interpret $\boldsymbol{\alpha} = (\alpha_T, \alpha_A, \alpha_V)$ as instance-level role signals: the modality with the largest weight is the *dominant* stream, and the remaining modalities serve as auxiliary evidence providers.

3.2 Role-Aware Fusion Module

Conditioned on $\boldsymbol{\alpha}$, the fusion module integrates multimodal cues by explicitly routing cross-modal interactions through the dominant stream, rather than applying uniform aggregation.

Tokenized Descriptors. Let $\mathbf{X}_m \in \mathbb{R}^{L_m \times H}$ be the sequence-like representation of modality m after projecting all modalities to a shared hidden size H (e.g., via linear adapters). Here L_T is the number of text tokens, while L_A and L_V are pseudo-token lengths for acoustic/visual descriptors.

Role-Conditioned Routing. We define the dominant modality index

$$m^\star = \arg\max_{m \in \mathcal{M}} \alpha_m, \tag{7}$$

and the auxiliary set $\mathcal{M}_{\neg m^\star} = \mathcal{M} \setminus \{m^\star\}$. The full distribution $\boldsymbol{\alpha}$ is retained for continuous weighting, while $m^\star$ specifies the interaction hub. Although arg max introduces a discrete selection, cross-modal evidence from auxiliary modalities is integrated through *continuous* weights α_m (Eq. 6), so near-tie cases (e.g., $\alpha_A = 0.49$ vs. $\alpha_V = 0.51$) yield only marginal changes in the fused representation when the hub identity swaps; the temperature τ additionally keeps the role distribution soft near decision boundaries.

Intra-Stream Refinement. We first apply self-attention to the dominant stream:

$$\mathbf{S}_{m^\star} = \mathrm{Attn}\Big(\mathbf{X}_{m^\star}\mathbf{W}^Q_{m^\star}, \mathbf{X}_{m^\star}\mathbf{W}^K_{m^\star}, \mathbf{X}_{m^\star}\mathbf{W}^V_{m^\star}\Big), \tag{8}$$

Cross-Modal Integration. For each auxiliary modality $m \in \mathcal{M}_{\neg m^\star}$, we compute dominant-query cross-attention:

$$\mathbf{C}_{m\to m^\star} = \mathrm{Attn}\Big(\mathbf{X}_{m^\star}\mathbf{W}^Q_{m^\star},\ \mathbf{X}_m\mathbf{W}^K_m,\ \mathbf{X}_m\mathbf{W}^V_m\Big), \tag{9}$$

and modulate it by the corresponding role weight:

$$\widehat{\mathbf{C}}_{m\to m^\star} = \alpha_m \cdot \mathbf{C}_{m\to m^\star}. \tag{10}$$

Here $\mathrm{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\big(\mathbf{Q}\mathbf{K}^\top/\sqrt{d}\big)\mathbf{V}$ (multi-head in practice).

Fusion Output and Decoding Interface. We concatenate the dominant stream, its refined state, and all weighted cross-modal evidence:

$$\mathbf{Z} = \Big[\mathbf{X}_{m^\star} \parallel \mathbf{S}_{m^\star} \parallel \{\widehat{\mathbf{C}}_{m\to m^\star}\}_{m\in\mathcal{M}_{\neg m^\star}}\Big], \tag{11}$$

and map it back to the decoder hidden size:

$$\mathbf{H}_{\text{fus}} = \text{LN}(\mathbf{Z}\mathbf{W}_{\text{fus}} + \mathbf{b}_{\text{fus}})\,. \tag{12}$$

To preserve textual grounding, we feed the decoder with text input embeddings augmented by $\mathbf{H}_{\text{fus}}$ (via concatenation followed by a lightweight projection), rather than allowing any non-text stream to supplant the textual backbone.

3.3 Emotion-Aware Decoding Unit

To couple response generation with affective modeling, we attach an Emotion-Aware Decoding Unit to decoder hidden states and optimize it jointly with language modeling. Let $\mathbf{h}_T \in \mathbb{R}^H$ denote the decoder hidden state at the last response position under teacher forcing.[2] We compute emotion logits:

$$\mathbf{z}^{(e)} = \mathbf{W}_e\,\text{LN}(\mathbf{h}_T) + \mathbf{b}_e, \quad \mathbf{p}^{(e)} = \text{softmax}(\mathbf{z}^{(e)}), \tag{13}$$

and define the auxiliary objective with cross-entropy:

$$\mathcal{L}_{\text{emo}} = -\log \mathbf{p}^{(e)}\Big[y^{(e)}\Big]\,. \tag{14}$$

The final training objective is

$$\mathcal{L} = \mathcal{L}_{\text{lm}} + \mathcal{L}_{\text{emo}}, \tag{15}$$

which promotes emotion-discriminative decoder representations and supports emotion-aligned response generation.

4 Experiments

4.1 Datasets

We evaluate on **MELD** [7] and **IEMOCAP** [2], two widely used multimodal emotion benchmarks with aligned text, audio, and visual streams and utterance-level emotion labels. MELD contains 13,708 utterances from 1,433 dialogues annotated with 7 emotion categories, and IEMOCAP comprises about 7,433 utterances from 151 dyadic sessions with 6 commonly-used emotion categories. For MELD, we follow the official train/dev/test split. For IEMOCAP, we adopt the standard session-based protocol (Sessions 1–4 for training, Session 5 for testing) and hold out a validation subset from the training sessions for model selection.

[2] In our implementation, $\mathbf{h}_T$ corresponds to `last_token_hidden_state`.

While leave-one-session-out cross-validation is the predominant protocol in the ERC literature [2], we opt for a fixed split primarily for two reasons: (1) it ensures a consistent comparison surface with MELD and with prior multimodal EDG work that also uses fixed train/test partitions [18]; and (2) IEMOCAP contains only five sessions, making each fold in a leave-one-out scheme a single session, which risks high evaluation variance on a test set of approximately 1,500 utterances and complicates checkpoint selection under heterogeneous session characteristics.

4.2 Implementation Details

All experiments run on an NVIDIA RTX 5090 GPU. Following ERGM [18], audio tracks are extracted with FFmpeg (16 kHz), word-aligned via the Montreal Forced Aligner, and each utterance is segmented into a synchronized audio–visual clip of 32 frames. We use CLIP ViT-L/14 and HuBERT-Large as frozen visual and acoustic encoders, respectively. All modalities are projected to a shared hidden size of $H = 768$ via linear adapters, and multi-head attention uses 12 heads across all fusion modules.

For RaMRA and the ERGM baseline, both based on a decoder-only GPT-style backbone (12 layers, 12 heads, $H = 768$), we fine-tune directly on MELD and IEMOCAP using the Adam optimizer with a learning rate of 1×10^{-5}, a batch size of 16, and 10 training epochs; generation uses nucleus sampling with $p = 0.8$. For LLM-based settings, we use the Qwen2.5-Omni Thinker module with a learning rate of 1×10^{-4}, feed the concatenated dialogue history as input, keep modality encoders frozen, and apply LoRA ($r = 16$, $\alpha = 32$). Identical decoding hyperparameters are used across all LLM-based models. Other baselines (CEM, SEEK, CASE, IAMM) follow the hyperparameters in their original papers, and all baselines are reproduced using their official codebases. The ERGM baseline is reproduced without the guidance-text branch (which is not released) and without EmpatheticDialogues pre-training, matching the input pipeline available to RaMRA.

4.3 Evaluation Metrics

We report the following automatic metrics: perplexity (PPL) for fluency, Dist-1/Dist-2 [5] for response-level lexical diversity, BERTScore [26] (F1) for semantic similarity with the ground-truth response, and emotion accuracy (**Emo Acc.**), computed by a fixed external emotion classifier—an MLP head trained on each dataset's utterance-level emotion labels following the standard ERC protocol—applied to the generated responses. All models are evaluated by the same external classifier. We acknowledge that unweighted average recall (UAR) is often preferred over accuracy for imbalanced emotion datasets. However, we report accuracy for two reasons: (1) all baselines we compare against [9, 13, 27, 23, 18] use accuracy as their primary emotion metric, making it the de facto standard in multimodal EDG; and (2) the class distributions of both MELD (7 categories, 4,710 train utterances) and IEMOCAP (6 categories, 5,810 train utterances) are

not severely skewed—the most and least frequent classes differ by a factor of approximately 3–4×, well within ranges where accuracy remains a reliable indicator. We additionally note that because all models are evaluated by the *same* external classifier under *identical* conditions, relative ranking under accuracy should closely mirror that under UAR.

Human Evaluation. We conduct a pairwise human study on 100 randomly sampled multimodal test instances from MELD and IEMOCAP (pooled for annotation). For each instance, three annotators with NLP research backgrounds compare two anonymized responses (order randomized) from RaMRA and the strongest baseline given the same multimodal inputs, and choose the better one under **Emp** (empathy/emotion alignment), **Coh** (contextual coherence), and **Flu** (fluency/naturalness). We report RaMRA's win rate per criterion. For LLM-based human evaluation, we use the same format-only prompt at inference for both ERGM(Qwen) and RaMRA(Qwen).

5 Results

5.1 Automatic Evaluation

Table 1. Main results on MELD and IEMOCAP. Lower is better for PPL; higher is better for Dist-1/2, Emo Acc., and BERTScore. Best results are in **bold**.

	MELD					IEMOCAP				
Model	**PPL↓**	**Dist-1↑**	**Dist-2↑**	**Emo Acc.↑**	**BERT↑**	**PPL↓**	**Dist-1↑**	**Dist-2↑**	**Emo Acc.↑**	**BERT↑**
Standard models										
CEM [9]	53.68	0.91	4.28	58.40	76.59	41.03	0.97	3.62	61.87	76.21
SEEK [13]	51.44	1.01	5.38	58.95	81.01	36.52	1.12	3.81	66.40	82.01
CASE [27]	34.02	1.10	5.81	60.48	80.73	29.51	1.16	4.11	65.94	82.44
IAMM [23]	33.94	1.08	6.55	58.72	82.99	30.68	1.42	5.09	66.78	83.10
ERGM [18]	27.63	1.98	9.87	66.45	84.98	22.34	**3.16**	10.83	72.05	83.18
RaMRA (Ours)	**24.45**	**1.99**	**10.51**	**70.44**	**85.05**	**16.60**	**3.16**	**11.11**	**77.93**	**83.32**
LLM-based models										
Llama 3.2–11B	18.36	3.06	20.03	60.18	81.01	12.26	3.95	21.89	60.52	78.12
ERGM (Qwen2.5-Omni) [20]	17.61	3.44	21.00	67.45	83.31	12.07	**3.99**	22.09	63.37	80.19
RaMRA (Qwen2.5-Omni)	**16.96**	**3.48**	**22.57**	**68.89**	**86.93**	**10.74**	3.96	**22.31**	**65.25**	**86.24**

Standard Models. As shown in Table 1, RaMRA consistently outperforms the strongest standard baseline ERGM on both MELD and IEMOCAP, with the most pronounced gains on fluency and affect alignment. On MELD, RaMRA reduces PPL from 27.63 to **24.45** and improves Emo Acc. from 66.45 to **70.44**, alongside a modest but consistent increase in BERTScore (84.98→85.05). Diversity is preserved at the compositional level (Dist-2: 9.87→**10.51**), while Dist-1 remains essentially unchanged (1.98→1.99), suggesting that role-conditioned routing amplifies informative cross-modal cues without introducing superficial lexical novelty.

On IEMOCAP, the gains are larger and better reflect RaMRA's motivation under heterogeneous modality salience: PPL drops substantially (22.34→**16.60**) and Emo Acc. increases by nearly six points (72.05→**77.93**). RaMRA also improves Dist-2 and BERTScore (10.83→**11.11**, 83.18→**83.32**), indicating better lexical composition and semantic fidelity. Dist-1 ties with ERGM (3.16 vs. **3.16**), which is consistent with our goal of controlled diversity rather than aggressive lexical perturbation.

LLM-Based Models. RaMRA remains beneficial for omni-modal LLMs. Compared with ERGM (Qwen2.5-Omni [20]), RaMRA (Qwen2.5-Omni) reduces PPL on both MELD and IEMOCAP (17.61→**16.96**; 12.07→**10.74**) and improves Emo Acc. (67.45→**68.89**; 63.37→**65.25**). Notably, RaMRA yields a substantial BERTScore gain, especially on IEMOCAP (80.19→**86.24**), suggesting that modality-role routing helps the LLM internalize multimodal evidence more effectively than uniform fusion; Dist-2 also increases on both datasets (21.00→**22.57**; 22.09→**22.31**). The tiny Dist-1 drop on IEMOCAP (3.99→3.96; absolute decrease of 0.03) is negligible given RaMRA's consistent gains in PPL, Emo Acc., and BERTScore. We select Qwen2.5-Omni and Llama 3.2–11B as representative open-source native multimodal LLMs because they support audio, vision, and text as native inputs, enabling fair, reproducible comparisons under identical fine-tuning conditions. While Llama 3.2–11B exhibits low PPL and high diversity, it lags behind Qwen-based omni-modal models in Emo Acc. and BERTScore, reinforcing that explicit audiovisual grounding is essential for affect-aware generation.

5.2 Human Evaluation

Table 2. Human preference win rate (%) of RaMRA over ERGM under three criteria: Emp (empathy/emotion alignment), Coh (contextual coherence), and Flu (fluency/naturalness).

Comparison	**Emp.↑**	**Coh↑**	**Flu↑**
RaMRA vs. ERGM	62.6	56.4	52.7
RaMRA (Qwen) vs. ERGM (Qwen)	54.1	57.1	60.8

Table 2 reports pairwise human preferences. Overall, RaMRA achieves higher win rates than ERGM across all three criteria (Emp/Coh/Flu). In the Standard setting, RaMRA's advantage is most pronounced on empathy alignment (62.6%), aligning with our motivation of amplifying the most informative affective modality. In the LLM-based setting (Qwen2.5-Omni), the win-rate profile shifts: the empathy gap narrows to 54.1% while fluency becomes the main beneficiary (60.8%). We attribute this to the strong affective prior already encoded

in Qwen2.5-Omni's large-scale multi-modal pre-training—when such an omni-modal LLM serves as the backbone, much of the speaker's emotion is already perceivable on its own, leaving RaMRA's role-conditioned routing to contribute mainly at the surface level of fluency and coherence.

5.3 Ablation Study

Table 3. Ablation results. MRD: Modality-Role Discriminator; RAF: Role-Aware Fusion Module; EDU: Emotion-Aware Decoding Unit. "w/o" removes the component; w/o MRD uses uniform role weights; w/o RAF replaces it with naive symmetric fusion; w/o EDU disables the auxiliary emotion supervision while keeping the rest of the pipeline intact. Emo Acc. is computed using a fixed external classifier on generated responses (independent of EDU, so Emo Acc. remains computable in all settings).

	MELD			**IEMOCAP**		
Setting	PPL↓	EmoAcc.↑	BERT↑	PPL↓	EmoAcc.↑	BERT↑
Full (RaMRA)	24.45	70.44	85.05	16.60	77.93	83.32
w/o MRD	26.81	68.72	83.79	17.98	75.16	82.88
w/o RAF	27.09	67.55	82.94	17.49	74.59	81.30
w/o EDU	25.63	62.80	84.55	17.07	72.89	82.24

Table 3 shows that each component of RaMRA is necessary for its overall gains. Removing the Modality-Role Discriminator (uniform roles) consistently degrades fluency and semantic fidelity (higher PPL and lower BERTScore), indicating the importance of instance-level role inference. Replacing role-aware routing with naive fusion yields the largest semantic drop, especially on IEMOCAP (BERTScore: 83.32→81.30) with clear Emo Acc. degradation, confirming that dominant-modality routing is more effective than uniform aggregation. Finally, removing the emotion-aware unit causes the largest loss in emotion consistency (Emo Acc. −7.64 on MELD; −5.04 on IEMOCAP), suggesting auxiliary affective supervision is critical for emotion-aligned generation.

6 Conclusion

We propose **RaMRA**, a framework that reframes multimodal EDG as instance-level modality-role assignment. By identifying a context-specific interaction hub and performing role-conditioned, asymmetric routing under textual grounding, RaMRA reduces redundant fusion and emphasizes the most reliable affective evidence. This principled role-based formulation, together with auxiliary emotion supervision, consistently improves emotion-aligned generation on MELD and IEMOCAP, demonstrating that adaptive multimodal affect perception is key to building emotionally intelligent dialogue systems.

Limitations and Future Work. While RaMRA demonstrates consistent gains, several limitations warrant discussion. **(1) Dataset scope.** Our evaluation is restricted to two dyadic English conversation datasets (MELD and IEMOCAP); validation on larger-scale, multi-party, or multilingual benchmarks would strengthen the generality claim. **(2) Encoder freezing.** We use frozen CLIP and HuBERT encoders, following the setup in ERGM [18]. Fine-tuning the acoustic encoder could potentially recover more affective signal from prosody, at the cost of additional computational overhead. **(3) Human evaluation scale.** Our human study involves 100 instances and three annotators, which is adequate for pairwise preference estimation but limits the statistical power for fine-grained per-category analysis; larger-scale annotations with more annotators would yield more robust subjective metrics. **(4) Modality role analysis.** The current work reports aggregated performance metrics but does not provide a breakdown of how often each modality is selected as dominant, or how role assignments correlate with ground-truth emotion categories. **(5) Robustness to degraded inputs.** We do not evaluate performance under missing or noisy modalities, a practical concern in real-world deployment. Future work will extend RaMRA along these axes, including per-emotion-category analysis, modality role visualizations, and evaluation under modality degradation scenarios.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Ai, W., Zhang, F., Shou, Y., Meng, T., Chen, H., Li, K.: Revisiting multimodal emotion recognition in conversation from the perspective of graph spectrum. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 39, pp. 11418–11426 (2025). https://doi.org/10.1609/aaai.v39i11.33242
2. Busso, C., Bulut, M., Lee, C.C., Kazemzadeh, A., Mower, E., Kim, S., Chang, J.N., Lee, S., Narayanan, S.S.: IEMOCAP: Interactive emotional dyadic motion capture database. In: Proceedings of the Sixth International Conference on Language Resources and Evaluation (LREC). European Language Resources Association (ELRA), Marrakech, Morocco (2008). https://doi.org/10.1007/s10579-008-9076-6
3. Fei, H., Zhang, H., Wang, B., Liao, L., Liu, Q., Cambria, E.: EmpathyEar: An open-source avatar multimodal empathetic chatbot. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations). pp. 61–71. Association for Computational Linguistics, Bangkok, Thailand (2024). https://doi.org/10.18653/v1/2024.acl-demos.7
4. Feng, X., Lin, Y., He, L., Li, Y., Chang, L., Zhou, Y.: Knowledge-guided dynamic modality attention fusion framework for multimodal sentiment analysis. In: Findings of the Association for Computational Linguistics: EMNLP 2024. pp. 14755–14766. Association for Computational Linguistics, Miami, Florida, USA (2024). https://doi.org/10.18653/v1/2024.findings-emnlp.865
5. Li, J., Galley, M., Brockett, C., Gao, J., Dolan, B.: A diversity-promoting objective function for neural conversation models. In: Proceedings of the 2016

Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT). pp. 110–119. Association for Computational Linguistics, San Diego, California (2016). https://doi.org/10.18653/v1/N16-1014
6. Liu, C., Xie, Z., Zhao, S., Zhou, J., Xu, T., Li, M., Chen, E.: Speak from heart: An emotion-guided LLM-based multimodal method for emotional dialogue generation. In: Proceedings of the 2024 International Conference on Multimedia Retrieval (ICMR). pp. 533–542. ACM, Phuket, Thailand (2024). https://doi.org/10.1145/3652583.3658104
7. Poria, S., Hazarika, D., Majumder, N., Naik, G., Cambria, E., Mihalcea, R.: MELD: A multimodal multi-party dataset for emotion recognition in conversations. In: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics. pp. 527–536. Association for Computational Linguistics, Florence, Italy (2019). https://doi.org/10.18653/v1/P19-1050
8. Qian, Y., Zhang, W., Liu, T.: Harnessing the power of large language models for empathetic response generation: Empirical investigations and improvements. In: Findings of the Association for Computational Linguistics: EMNLP 2023. pp. 6516–6528. Association for Computational Linguistics, Singapore (2023). https://doi.org/10.18653/v1/2023.findings-emnlp.433
9. Sabour, S., Zheng, C., Huang, M.: CEM: Commonsense-aware empathetic response generation. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 36, pp. 11229–11237. AAAI Press, Virtual Conference (2022). https://doi.org/10.1609/aaai.v36i10.21373
10. Shi, T., Huang, S.L.: MultiEMO: An attention-based correlation-aware multimodal fusion framework for emotion recognition in conversations. In: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1). pp. 14752–14766. Association for Computational Linguistics, Toronto, Canada (2023). https://doi.org/10.18653/v1/2023.acl-long.824
11. Tu, G., Xie, T., Liang, B., Wang, H., Xu, R.: Adaptive graph learning for multimodal conversational emotion detection. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 38, pp. 19089–19097 (2024). https://doi.org/10.1609/aaai.v38i17.29876
12. Wang, H., Zhang, G., Chen, J., Li, J., Wang, Y., Guo, Y.: Empathy omni: Enabling empathetic speech response generation through large language models. arXiv preprint arXiv:2508.18655 (2025). https://doi.org/10.48550/arXiv.2508.18655
13. Wang, L., Li, J., Lin, Z., Meng, F., Yang, C., Wang, W., Zhou, J.: Empathetic dialogue generation via sensitive emotion recognition and sensible knowledge selection. In: Findings of the Association for Computational Linguistics: EMNLP 2022. pp. 4634–4645. Association for Computational Linguistics, Abu Dhabi, United Arab Emirates (2022). https://doi.org/10.18653/v1/2022.findings-emnlp.340
14. Wang, Y., He, J., Wang, D., Wang, Q., Wan, B., Luo, X.: Multimodal transformer with adaptive modality weighting for multimodal sentiment analysis. Neurocomputing **572**, 127181 (2024). https://doi.org/10.1016/j.neucom.2023.127181
15. Weng, Z., Wu, Z., Li, H., Chen, J., Jiang, Y.G.: HCMS: Hierarchical and conditional modality selection for efficient video recognition. ACM Transactions on Multimedia Computing, Communications, and Applications **20**(2), 1–18 (2023). https://doi.org/10.1145/3572776
16. Wu, C., Cai, Y., Liu, Y., Zhu, P., Xue, Y., Gong, Z., Hirschberg, J., Ma, B.: Multimodal emotion recognition in conversations: A survey of methods, trends, challenges and prospects. In: Findings of the Association for Computational Lin-

guistics: EMNLP 2025. pp. 6257–6274. Association for Computational Linguistics, Suzhou, China (2025). https://doi.org/10.18653/v1/2025.findings-emnlp.332
17. Wu, J., Huang, X., Zhu, Z., Wang, S.: From traits to empathy: Personality-aware multimodal empathetic response generation. In: Proceedings of the 31st International Conference on Computational Linguistics. pp. 8925–8938. Association for Computational Linguistics, Abu Dhabi, UAE (2025), https://aclanthology.org/2025.coling-main.598/
18. Wu, J., Wang, S., Chang, Y., Zhu, Z.: Empathetic response generation through multi-modality. IEEE Transactions on Affective Computing **16**(4), 3614–3623 (2025). https://doi.org/10.1109/TAFFC.2025.3599869
19. Wu, Z., Gong, Z., Koo, J., Hirschberg, J.: Multimodal multi-loss fusion network for sentiment analysis. In: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL). pp. 3588–3602. Association for Computational Linguistics, Mexico City, Mexico (2024). https://doi.org/10.18653/v1/2024.naacl-long.197
20. Xu, J., Guo, Z., He, J., Hu, H., He, T., Bai, S., Chen, K., Wang, J., Fan, Y., Dang, K., Zhang, B., Wang, X., Chu, Y., Lin, J.: Qwen2.5-omni technical report. arXiv preprint arXiv:2503.20215 (2025). https://doi.org/10.48550/arXiv.2503.20215
21. Xue, Z., Marculescu, R.: Dynamic multimodal fusion. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops. pp. 2575–2584 (2023). https://doi.org/10.1109/CVPRW59228.2023.00256
22. Yan, H., Zhu, Y., Zheng, K., Liu, B., Cao, H., Jiang, D., Xu, L.: Talk with human-like agents: Empathetic dialogue through perceptible acoustic reception and reaction. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1). pp. 15009–15022. Association for Computational Linguistics, Bangkok, Thailand (2024). https://doi.org/10.18653/v1/2024.acl-long.801
23. Yang, Z., Ren, Z., Wang, Y., Sun, H., Chen, C., Zhu, X., Liao, X.: An iterative associative memory model for empathetic response generation. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1). pp. 3081–3092. Association for Computational Linguistics, Bangkok, Thailand (2024). https://doi.org/10.18653/v1/2024.acl-long.170
24. Zhang, D., Nayak, R., Bashar, M.A.: Pre-gating and contextual attention gate — a new fusion method for multi-modal data tasks. Neural Networks **179**, 106553 (2024). https://doi.org/10.1016/j.neunet.2024.106553
25. Zhang, H., Meng, Z., Luo, M., Han, H., Liao, L., Cambria, E., Fei, H.: Towards multimodal empathetic response generation: A rich text-speech-vision avatar-based benchmark. In: Proceedings of the ACM Web Conference 2025. pp. 2872–2881. ACM, Sydney, Australia (2025). https://doi.org/10.1145/3696410.3714739
26. Zhang, T., Kishore, V., Wu, F., Weinberger, K.Q., Artzi, Y.: BERTScore: Evaluating text generation with BERT. In: Proceedings of the 8th International Conference on Learning Representations. OpenReview.net, Addis Ababa, Ethiopia (2020)
27. Zhou, J., Zheng, C., Wang, B., Zhang, Z., Huang, M.: CASE: Aligning coarse-to-fine cognition and affection for empathetic response generation. In: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1). pp. 8223–8237. Association for Computational Linguistics, Toronto, Canada (2023). https://doi.org/10.18653/v1/2023.acl-long.457